MIL-C-49228(CR)

16 June 1981

PRODUCTION SPECIFICATION

COMMUNICATIONS CENTRAL AN/ASC-15( )(V)1,2

This specification is approved for use by the US Army Communications Electronics Command and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for manufacture and acceptance of a Rack Equipment Console (see 6.1) with associated installation kit.

1.2 Appendices. The appendices listed below form a part of this specification:

Appendix A - Major Component List Comprising Communications Central AN/ASC-15( )(V)1.

Appendix B - Major Component List Comprising Communications Central AN/ASC-15( )(V)2.

Appendix C - TEMPEST Requirements (Title Unclassified, Document Confidential).

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of the specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications Research and Development Command, ATTN: DRDCO-CM, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5895

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SPECIFICATIONS

Military

MIL-E-1	Electronic Devices
MIL-P-116	Preservation-Packaging, Methods of
MIL-C-172	Case; Bases; Mounting; and Mounts, Vibration
MIL-T-713	Cable Harnessing
MIL-B-5087	Bonding, Electrical and Lighting Protection for Aerospace Systems
MIL-W-5088	Wiring Aircraft, Selection and Installation of
MIL-E-5400	Electronic Equipment, Airborne, General Specification
MIL-T-5422	Testing Environmental, Aircraft Electronic Equipment
MIL-C-5809	Circuit Breakers, Trip Free, Aircraft General Specification for (ASG)
MIL-C-6499	Clock Specification
MIL-C-6781	Control-Panel, Aircraft Equipment Rack and Console Mounted
MIL-P-7788	Plate, Plastic, Lighting Panels, Integrally Illuminated
MIL-S-8834	Switches, Toggle, Positive Break, Aircraft General Specification for
MIL-M-10578	Acid Cleaning
MIL-M-13231	Markings
MIL-P-13949	Printed Wiring Board Lammates
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-S-19500	Tubes, Electrons, and Semiconductors
MIL-C-27715	Circuit Breaker, Trip Free, High Temperature, Aircraft, General Specification for
MIL-C-39012	Connectors, Coaxial, Radio Frequency General Specification
MIL-W-45205A	Structural Welding
MIL-I-46058	Conformal Coatings
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-P-55110	Printed Wiring Boards
MIL-C-55338	Communications Security Requirements for Avionics Communication Systems, General Specification for

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MIL-C-55338/3	TEMPEST Limits for Radio Set AN/ARC-131
MIL-S-55556	Specification, Preparation of Installation and Acceptance Testing of Electronics Equipment in Aircraft
SPRAGUE #JN17-3346A	Power Line Filter

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling for Inspection
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-252	Inspection Techniques
MIL-STD-275	Printed Wiring Boards
MIL-STD-454F	Standard General Requirements for Electronic Equipment
MIL-STD-461A	Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (For System and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781C	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-877	Antenna Subsystems, Airborne Criteria for Design and Location

SPI 1G00051
National Security Agency

NACSEM 5100	Compromising Emanations Laboratory Test Standards, Electromagnetics (Title Unclassified; Document Confidential)

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DRAWINGS

CERCOM-DL-972560
CERCOM-DL-972723
CERCOM-DL-972728
(including latest changes with all product improvements)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Rack Equipment Console, hereinafter referred to as console, is an enclosed frame with integral wiring that is used in US Army aircraft. It operates from a nominal voltage of 27.5 volts dc derived from the aircraft power systems. The console when configured with the equipments shown in the appendices functions as an airborne command post, providing tactical voice communication in both secure and nonsecure modes. Also, the console functions as a secure automatic communication retransmission station. The console provides intercommunication facilities for up to six users and communications management for up to three operators. It is further designed to be rapidly installed or removed from the aircraft. The installation kit makes the console operational in the aircraft.

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3.2 First Article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.4).

3.2.1 When specified, the contractor shall provide sample units for first article packaging inspection and approval (see 4.2.1 and 6.7.1).

3.3 Construction. The console shall be constructed in accordance with the following applicable drawing and data lists as determined by the contracting officer:

CERCOM-DL-972560 - Rack Equipment Console (less equipment)

- Aircraft Installation Kit

3.4 Parts, Materials and Processes (See 4.5). In addition to the requirements of this specification, parts, materials and processes shall be in accordance with Specification MIL-E-5400, including the selection requirements therein.

3.4.1 Printed Wiring Assemblies. Printed wiring assemblies shall be processed for compliance with the requirements specified, in order to affect life and service ability of the assemblies.

3.4.1.1 Printed Wiring Boards. Printed wiring boards used in the construction of printed wiring assemblies, shall be a product which has passed the requirements of MIL-P-55110, MIL-STD-275 and the master drawing. However, those parts of MIL-STD-275 which are in conflict with the applicable printed wiring board drawings and which require redesign of the printed wiring boards shall not apply. Conflicts between master drawings and MIL-STD-275 need not be submitted as required by MIL-P-55110. The printed wiring assemblies shall be clean and free from dirt, oil, corrosion, corrosion products, salt, smut, grease, fingerprints and all other foreign matter. There shall be no whiskers or solder or plating left on the conductor pattern. Printed wiring assemblies shall be processed in such a manner as to meet the requirements specified, to be uniform in quality and to be free from defects that will affect life and serviceability.

3.4.1.2 Laminate. The laminate shall be type GB or GE in accordance with MIL-P-13949, unless otherwise specified in the master drawing.

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3.4.1.3 Plating. Plating of printed wiring boards shall be in accordance with MIL-P-55110, and the master drawing.

3.4.1.4 Protective Coating. Printed wiring assemblies shall be protected on both sides of the printed wiring assembly with an epoxy-type conformal coating to protect the pattern against moisture, dust and other contaminants. The coating shall be in accordance with MIL-I-46058. The thickness of the coating shall be .001 IN. MIN, TO .008 IN. MAX. Assemblies having adjustable components shall not have the adjustable portion covered with the coating. Electrical and mechanical mating surfaces such as connector contact points, test points, screw threads, bearing surfaces, etc, shall not be coated. Hermetic sealing of the assembly in a dry atmosphere shall be acceptable in place of conformal coating.

3.4.1.5 Solderability. Solderability of printed wiring boards shall conform to MIL-P-55110.

3.4.2 Wiring and Cabling. Wiring and cabling shall be neat and sturdy and meet the requirements of MIL-W-5088.

3.4.2.1 Cabling. Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling or wires shall be affected by tape or twin conforming to MIL-T-713. Individual conductors thus secured lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.4.2.2 Slack. Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

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c. To provide for at least two replacements of the part to which the wire or cable is connected, except for certain radio-frequency leads which must be as short as possible for electrical reasons.

d. To facilitate field repair of broken or cut wires.

e. To prevent chafing or breaking of individual wires due to repeated flexing hinged parts.

3.4.2.3 Protection. Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.4.2.4 Support. Wires and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner.

3.4.2.5 Clearance. Clearance between wires or cables, and parts such as electron tubes and resistors, shall be provided to avoid determination of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.4.2.6 Splicing. Wires in continuous run between two terminals shall not be spliced during the wiring operation.

3.4.2.7 Connections.

3.4.2.7.1 General. Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength or solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but

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not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and non-metallic material.

3.4.2.7.2 Bonds and Grounds. The console shall be constructed with adequate provisions for bonding the console to the airframe structure and achieving a 2.5 milliohm or less bonding impedance. Bonding shall not be accomplished through bolts connecting the console to the floor. Shock mounted equipments or those employing vibration isolators shall utilize bonding straps to bypass the mount and achieve the 2.5 milliohm bonding impedance. Bonding jumpers shall be of the solid metal type and shall be as short as possible. In no case, shall the length to width ratio of the jumper be in excess of 5 to 1. Surfaces being bonded together shall be prepared by removing all anodic film, grease, paint, lacquer or other foreign and high resistance materials or agents from the immediate area of the bond to insure negligible radio frequency (RF) impedance. Bonding shall be in accordance with MIL-B-5087.

3.4.3 Controls. All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition. Circuit breakers shall meet the requirements of MIL-C-5809, switches shall meet the requirements of MIL-S-8834 and MIL-C-27715.

3.4.4 Tubes, Electrons and Semiconductors. These electron devices (tubes and semiconductors) shall conform to MIL-E-1 and MIL-S-19500.

3.4.5 Tropicalization of Materials. Materials shall be treated as follows:

3.4.5.1 Treating Materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating compounds shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.4.5.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

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3.4.6 Cleaning.

3.4.6.1 Parts. After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454F, or as specified in an applicable document.

3.4.6.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.4.7 Welding. Welding shall be in accordance with the equipment drawing. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. When spot or other intermittent welds are used to hold a part, the numbers of welds shall be at least two. In addition, all welds shall be in accordance with MIL-W-45205A class A.

3.4.7.1 Cleaning Prior to Welding. Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.4.7.2 Process. Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum or magnesium.

3.4.7.3 Cleaning After Welding. Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.4.8 Riveting. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

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3.4.9 Securing of Parts.

3.4.9.1 General. Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.4.9.2 Securing of Parts by Threaded Fasteners. Nylok or other screws with plastic devices, and Loctite or similar sealants shall not be used.

3.4.9.3 Equipment Mounts. All radios, encryption devices and repeaters shall be shock mounted and meet the requirements of MIL-C-172. This paragraph does not apply to the AN/ARC-164 radio set.

3.4.10 Soldering.

3.4.10.1 Solder. Solder shall conform to the equipment drawings.

3.4.10.2 Acid or Acid Salts. No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.4.10.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.5 Reliability Burn-In/Debugging (See 4.7). Each console on order shall be subjected to a 48 hour burn-in period with the final 24 hours failure free (see 6.6).

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3.6 Electrical Requirements (See 4.9). Unless otherwise specified, the following requirements shall apply to the console under the standard conditions as defined in 4.8.

3.6.1 Sensitivity. RF signal inputs of 0.5 microvolts with a carrier frequency of 30 MHz, 42 MHz, 52 MHz, 53 MHz, 67 MHz and 75 MHz, frequency modulated at 1000 Hz ± 8 kHz deviation, when applied to antenna inputs FM-1, FM-2 or FM-3 shall produce a signal plus noise plus distortion to noise plus distortion ratio (SINAD) of 10 dB minimum when measured at a level of 50 millivolts at the FM-1 audio output terminal TB-8 pin 8, FM-2 audio output terminal TB-9 pin 8, and FM-3 audio output terminal TB-8 pin 16, respectively. The SINAD will then be measured at all of the six headphone outputs. Not more than a 2 microvolt increase in the RF signal level shall be required to obtain the same 10 dB audio SINAD.

3.6.2 When the AN/ARC-164 or AN/ARC-51BX radio set is used in place of the AN/ARC-131 in the FM-3 position, then the sensitivity shall be as follows: An RF signal of 5 microvolts with a carrier frequency of 225 MHz, 245 MHz, 268 MHz, 288.850 MHz, 312 MHz, 312.050 MHz, 377.750 MHz and 399 MHz, amplitude modulated 30% with a 1000 Hz tone, when applied to the UHF/VHF antenna input shall produce a minimum SINAD of 10 dB when measured at terminal TB-10 pin 8. This measurement will then be made at all of the six headphone outputs and not more than a 2 microvolt increase in signal level will be permitted to obtain the same 10 dB SINAD (see Fig 2).

3.6.3 Transmitter Power Output: With FM-1, FM-2 and FM-3 antenna ports each connected to a 50 ohm load the AN/ARC-131 transmitters shall produce a minimum of 10 watts RF power into its respective 50 ohm load when keyed from any of the U-94 A/U headset-microphone cords. This shall include all transmitter frequencies from 30 MHz to 75.95 Hz in 50 kHz steps. The XMIT Hi-Lo switch on the AN/ARC-131 shall reduce the power output to 0.5 watt minimum to 1.5 watts maximum.

3.6.3.1 When the AN/ARC-51BX radio set is used in place of the AN/ARC-131 in the FM-3 position, the power output at the UHF/VHF antenna port shall be greater than 16 watts RF power into a 50 ohm load from 225 MHz to 399.975 MHz. The transmitter will be keyed from any of the six stations using the U-94 A/U headset-microphone cords with the C-1611/AIC transmit control set to the #4 position (see Fig 2).

3.6.3.2 When the AN/ARC-164 radio set is used in the FM-3 position, the transmitter power output when measured into a 50 ohm load at the UHF/VHF antenna port shall read 9 watts minimum from 225 MHz to 399.975 MHz. The C-1611 shall be set to the #4 position.

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3.6.4 Sidetone. When a transmitter is keyed and the operator is talking into the microphone from any one of the 6 stations, sidetone shall be heard in the appropriate headset. Caution should be taken to assure the antenna output ports, FM-1, FM-2 and FM-3, are properly loaded with 50 ohms.

3.6.5 Retransmission. When the console is connected into the retransmission configuration, the amount of received signal power required to obtain a 10 dB audio SINAD at a receiver site shall be noted for the clear text mode. The system will then be switched to the cypher mode. A reduction of 4 dB received signal power is allowed to obtain the same 10 dB SINAD at the receiver site in the cypher text mode (see Fig 1).

3.7 Service Conditions (See 4.12). The console shall meet the environmental service conditions requirement as specified in this specification.

3.7.1 Temperature (See 4.12.1). The console and/or associated installation kit shall operate with no degradation of performance at ambient temperatures ranging from -54 C to +55 C. In addition, the console shall withstand ambient storage temperatures from -62 C to +85 C with no damage.

3.7.2 Altitude (See 4.12.1). The console and/or associated installation kit shall operate normally with no damage at altitudes ranging from sea level to 30,000 feet above sea level with no degradation of performance.

3.7.3 Humidity (See 4.12.2.1). The console shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.7.4 Rain. The console and/or associated installation kit shall meet the rain environment given in paragraph 4.12.2.2.

3.7.5 Fungus (See 4.12.3). The console and/or associated installation kit, in the assembled and ready for delivery condition, shall provide no nutrients in material coating, or contamination form or support fungal growth when tested in accordance with paragraph 4.12.3. In addition, the console shall operate satisfactorily after exposure to the test conditions of paragraph 4.12.3. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454F, shall be used.

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3.7.6 Dust (See 4.12.6). The console shall withstand, in both operating and non-operating condition, exposure to fine dust particles with wind speeds of 35 Knots.

3.7.7 Salt Atmosphere (See 4.12.4). The console and/or associated installation kit shall meet the salt-sea atmosphere requirements.

3.7.8 Explosive Conditions (See 4.12.5). The equipment shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere.

3.7.9 Vibration (See 4.12.8).

3.7.9.1 Equipment Normally Mounted. When normally mounted (with vibration isolators in place, if any), the console shall operate satisfactorily when subjected to vibration within the frequency range and amplitude as shown on Figure 514.2-3 of MIL-STD-810C for the applicable equipment category.

3.7.9.2 Equipment with Isolators Removed. The console and/or its associated installation kit normally mounted on isolators shall operate satisfactorily with isolators removed when subjected to vibration within the frequency range and amplitude as shown on Figure 514.2-3 of MIL-STD-810C.

3.7.10 Shock (See 4.12.7).

3.7.10.1 Equipment. The console and/or associated installation kit (with vibration isolators in place, if any), shall not suffer damage or subsequently fail to provide specified performance when subjected to 18 impact shocks of 15 g, consisting of 3 shocks in opposite directions along each of 3 mutually perpendicular axes, each shock impulse shall have a time duration of 11 $\pm$ 1 milliseconds. The "g" value shall be within $\pm$ 10 percent when measured with a 0.2 to 250 Hz filter, and maximum "g" value shall occur at approximately 5-1/2 milliseconds.

3.7.10.2 Mounting Base (Crash Safety). With excursion stops or bumpers in place and with maximum rated load applied in a normal manner,

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the mounting base, individual isolators, or other attaching devices shall withstand at least 12 impact shocks of 30 g, consisting of 2 shocks in opposite directions along each of 3 mutually perpendicular axes. Each shock impulse shall have a time duration of 11 + 1 milliseconds. The "g" value shall be within + 10 percent when measured with a 0.2 to 250 Hz filter, and maximum "g" value shall occur at approximately 5-1/2 milliseconds. Bending and distortion is permitted; however, there shall be no failure of the attaching joints and the equipment or dummy load shall remain in place (see 4.12.7.2).

3.7.11 Bench Handling (See 4.12.7.3). The console and/or associated installation kit shall withstand the shock environment encountered during servicing without degradation of performance.

3.8 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454F, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (see 4.14).

3.8.1 Edge Rounding, Exposed. Exposed edges and corners shall be adequately rounded to preclude laceration and puncture hazards.

3.8.2 Radioactive Materials. Radioactive materials shall not be used (e.g., luminous dials/marking, electron tubes, surge arrestors) except as specified in MIL-C-6499 for the console clock. The console shall bear proper warning label indicating if clock dial is radioactive.

3.9 Electromagnetic Interference (See 4.11). The console with all equipments installed shall meet the emission and susceptibility requirements of MIL-STD-461A, Notice 4. The following requirements of that standard are applicable:

CE01	CS01	(a) RE02	(b) RS03
(a) CE04	CS02	RE02.1	
	CS06		

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a. Conducted switching transients shall be measured at 0.014, 0.800, 8.0 and 25.0 MHz, and the levels compared to the limits of Figure 5 of MIL-STD-461A, Notice 4. Radiated switching transients shall be measured at 0.014, 0.200, 1.2, 14.8 and 400 MHz, and compared to the limits of Figure 18 of the military standard.

b. The console and its components shall not malfunction, exhibit degraded performance or undesirable response when immersed in electric fields of the following intensity using the modulation characteristics specified.

Frequency Range	Field Intensity	Modulation Characteristics
0.010 to 1.9 MHz	1 volt/meter	AM, 50%, 1000 Hz tone
2 to 29.9 MHz	10 volts/meter	AM, 50%, 1000 Hz tone
30 to 75.9 MHz	10 volts/meter	FM, ± 8 kHz deviation, 1000 Hz tone
76 to 399.9 MHz	10 volts/meter	AM, 50%, 1000 Hz tone
400 to 1999.9 MHz	10 volts/meter	Pulse, 0.1 usec pulse width, 400 pps
2 to 12.4 GHz	5 volts/meter	Pulse, 0.1 usec pulse width, 400 pps

3.10 TEMPEST (See 4.10). The communications console shall meet the requirements of Appendix C.

3.11 Panel and Indicator Lights. All control panels shall utilize aircraft instrument panel type lighting to facilitate operation in a darkened area in accordance with MIL-P-7788. All indicator lights shall be so wired as to permit checking the condition of each filament with a "press to check" operation.

3.11.1 The console shall have a red panel light indicating primary power to the console. In addition, the lamp lens shall have a dimming capability. The lamp shall operate from 28Vdc.

3.11.2 The console shall have three green "cypher" indicator lights so operators are able to determine the mode of operation of the encryption devices. There shall be one green panel light for each receiver-transmitter (3 total). In addition, the lamp lens shall have a dimming capability. The lamp shall operate from 28Vdc.

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3.11.3 Clock. Each console shall be configured with an 8-day mechanical clock meeting the requirements of MIL-C-6499. In addition, the clock face shall have phosphorescent numbers and the clock hands shall be phosphorescent. The clock shall be securely attached to the console and readily visible to all three operators. The clock shall be easily removed from the console with no tools required.

3.12 Power Line Filters. Each radio set shall have a dc power line filter whose requirements shall be at least that of the Sprague power line filter (28Vdc, 15 amps #JN17-3346A).

3.13 Finish, Protective (See 4.14). Unit(s) shall be given protective finish in accordance with MIL-F-14072 and the unit(s) drawings. This includes finish of hardware such as hinges, screws, etc, and necessary touch-up after mounting. The final paint film shall be Color No. 37038 (Lusterless Aviation Black) in accordance with FED-STD-595. Wrinkle finish shall not be used.

3.14 Marking (See 4.14). Marking shall be in conformance with MIL-M-13231. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly.

3.14.1 Serial Numbers. All consoles on order shall be consecutively serial numbered.

3.14.2 Caution Plate. Each console shall have a safety warning plate riveted to each side. The two plates shall be 3" x 5" with black letters on a yellow background. The legend shall be as follows:

CAUTION

4 PERSON LIFT

245 LBS. TOTAL WEIGHT

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3.14.2.1 Warning Plate. Each console shall have a 5 in. x 7 in. warning plate riveted to the junction box cover JB1 stating:

WARNING

This console is not a secure means of communications unless all of its associated RT's are equipped with COMSEC security devices. When operating with less than a full complement of COMSEC equipment DO NOT simultaneously transmit on a non-secure RT while classified information is being discussed on any other channel of this console, including the associated intercom system.

The plate shall have a standard red and white background with black letters. In addition, the plate shall be aluminum .025 in. thick minimum. The letters shall be maximum size as permitted by the size of the plate.

3.14.3 Nameplates. Nameplates shall be in accordance with MIL-STD-130.

3.15 Interchangeability. Interchangeability shall exist between all units and replaceable assemblies, subassemblies, and parts for all equipment delivered on the contract in accordance with MIL-STD-454F, Requirement 7. In addition, no part or piece shall be unique. All panels, hardware, radios, encryption devices shall be interchangeable with its counterpart on any other AN/ASC-15( )(V) console (see 4.15).

3.15.1 Interchangeability conflicts. Should any conflicts arise between any of the requirements of this specification or the detail specification and the requirement for interchangeability, the contractor should immediately inform the procuring service of such conflict. Unless advised otherwise, the interchangeability requirement shall govern.

3.16 Packaging (Rough Handling Test). When rough handling test in accordance with MIL-P-116 is required by the contract, an operational inspection shall be conducted to determine freedom from operational malfunction caused by rough handling (see 3.6 and 4.6).

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3.17 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of MIL-STD-454F, equipment drawings and the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspections. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification Inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.6).

4.2.1 Quality Conformance Inspection. Quality conformance inspection of Section 3 requirements shall be in accordance with the Group A, Group B, when required see 6.4, the Group C and the rough handling requirements of MIL-P-116. Inspection lots for packaging shall be in accordance with MIL-P-116.

4.3 Inspection Conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified herein (see 4.8).

4.4 First Article (Preproduction) Inspection. First article inspections shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

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4.4.1 Sample Size. Four Rack Equipment Consoles shall be subjected to first article inspection.

4.4.2 First Article Inspection. This inspection shall consist of the inspection specified in subsidiary documents covering the item listed in para 4.5, the inspection specified for Group A, Group B and Group C (see Table II, III and IV respectively) and the inspection specified in para 4.7. The inspection of First Article samples, identified as Unit 1, Unit 2, Unit 3, and Unit 4, shall be performed in the order shown in Table I.

TABLE I - ORDER OF FIRST ARTICLE INSPECTION

Performance/Electrical (Para 3.17, 4.9)	Unit I	Unit II	Unit III	Unit IV
Temperature/Altitude	1			
Explosive Atmosphere			1	
Dust		1		
Vibration		2		
Humidity	2			
Fungus			2	
Salt Atmosphere	3			
Rain	4			
Shock		3		
Electromagnetic Interference				1
TEMPEST				2

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

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4.4.4 Test Plan. The contractor shall prepare and submit a test plan and test specification for review and approval by the contracting officer. The test plan shall be delivered in accordance with the approved DD Form 1423 incorporated in the contract. The test plan is to be followed in the performance of the first article testing and it shall outline all test devices, test facilities, methods, block diagrams for setups, procedures, sequences, extent of test data to be obtained and measurement values which shall be obtained together with permissable tolerances. Electronic test equipment to be used shall be identified by manufacturer and type, or by military identification. When tests call for the use of special contractor-built test harnesses and test devices; schematics, drawings, and descriptions shall be furnished. The test plans shall be submitted prior to the date the contractor has scheduled presubmission tests. The procuring activity shall, after the test plan is received, furnish a statement of approval or a statement of disapproval accompanied by an explicit review of the deficiencies of the test plan. When a test plan is disapproved, the contractor shall prepare a new test plan and submit 3 copies for approval by the contracting officer's representative.

4.5 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection during production and before preparation for delivery:

Para. Requirement	Item
3.4	Parts, Materials and Processes
3.11	Lamp Assembly
3.13	Finish
3.14	Marking
3.17	Workmanship

4.6 QUALITY CONFORMANCE INSPECTION

4.6.1 Inspection of Product for Delivery. Inspection of product for delivery shall consist of groups A, B and C inspection. This does not relieve the contractor of his responsibility for performing any

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additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.6.1.1 Group A Inspection. Group A inspection shall consist of the inspections specified in Table II, in any order which is satisfactory to the Government, except that visual and mechanical inspection shall be first.

4.6.1.1.1 Sampling Plan. Each unit produced shall be inspected for conformance to all the inspection requirements of Table II. Each lot shall then be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels. The acceptable quality level (AQL) shall be as specified in Table II. Major and minor defects shall be as defined in MIL-STD-105.

4.6.1.1.2 Rejected Lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for inspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

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TABLE II - GROUP A INSPECTION

Test/Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical				
Soldering	3.4.10	4.14	1%	4%
Workmanship	3.17	4.14	1%	4%
Fastener Hardware	3.4.9	4.14	1%	4%
Structural Welding	3.4.7	4.14	1%	4%
Finish	3.13	4.14	1%	4%
Markings	3.14	4.14	1%	4%
Safety	3.8, 3.8.1, 3.8.2	4.14	1%	4%
Electrical				
Sensitivity	3.6.1, 3.6.2	4.9	1% for the entire electrical inspection	
Transmitter Power Output	3.6.3	4.9		
Sidetone	3.6.4	4.9		
Retransmission	3.6.5	4.9		

4.6.1.2 Group B Inspection. Group B inspection shall consist of the inspection specified in Table III, and the sample shall be selected from inspection lots that have passed group A inspection.

4.6.1.2.1 Sampling Plan. The sampling plan shall be in accordance with MIL-STD-105 and the special procedures for small sample inspection contained therein. Inspection level shall be S-4 for normal inspection and S-3 for reduced inspection. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 6.5 percent defective.

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4.6.1.2.2 Rejected Lots. If an inspection lot is rejected the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for inspection. Resubmitted lots shall be isnpected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE III - GROUP B INSPECTION

Inspection	Req Para	Insp Para
Interchangeability	3.15	4.15

4.6.1.2.3 Disposition of Sample Units. Sample units which have passed the group B inspection may be delivered on the contract, if the lot is accepted and the sample units are still within specified tolerances.

4.6.2 Periodic Inspection. Periodic inspection shall consist of groups A and B. Except where the results of these inspections show noncompliance with the applicable requirements (see 6.5), delivery of products which have passed groups A, B and C shall not be delayed pending the results of these periodic inspections.

4.6.2.1 Group C Inspection. Group C inspection shall consist of the inspections specified in Table IV, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

4.6.2.1.1 Sampling Plan. Units shall be selected at random for each group of inspections in Table IV as specified in 4.6.2.1.2 through 4.6.2.1.5. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lots.

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TABLE IV - GROUP C INSPECTION

Inspection	Req Para	Insp Para
Subgroup I		
Temperature/Altitude	3.7.1/3.7.2	4.12.1
Humidity	3.7.3	4.12.2.1
Vibration	3.7.9	4.12.8
Bench Handling	3.7.11	4.12.7.3
Equipment Shock	3.7.10.1	4.12.7
Crash Safety Shock	3.7.10.2	4.12.7.2
Subgroup II		
Fungus	3.7.5	4.12.3
Salt Atmosphere	3.7.7	4.12.4
Explosive Atmosphere	3.7.8	4.12.5
Dust	3.7.6	4.12.6
Rain	3.7.4	4.12.2.2
Subgroup III		
Electromagnetic Interference	3.9	4.11
Subgroup IV		
TEMPEST	3.10	4.10

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4.6.2.1.2 Subgroup I. For this subgroup, two units shall be selected from the first forty (40) production units or 60 days whichever occurs first. For subsequent inspection, two (2) units from every 120 days or 120 production units whichever occurs first. The test in this subgroup may be divided between the two units, with the tests on the first unit being Temperature/Altitude, Humidity, and on the second unit being Vibration and Shocks.

4.6.2.1.3 Subgroup II. For this subgroup, two (2) units shall be selected from the first forty (40) production units or 60 days whichever occurs first. For subsequent inspections, two units shall be selected every 180 days or 180 production units whichever occurs first. The tests in this subgroup may be divided between the two units, with the tests on the first unit being Rain, Dust, and Explosive, on the second unit being Fungus and Salt.

4.6.2.1.4 Subgroup III. For this subgroup, one production unit selected from first, midway, and final production lot through the production.

4.6.2.1.5 Subgroup IV. For this subgroup one (1) production unit shall be selected midway through the production lot (see 4.10.2).

4.6.2.1.6 Failures. Actions required relative to Group C failures shall be as specified in the contract.

4.6.2.1.7 Disposition of Sample Units. Sample units which have been subjected to and passed Group C inspection may be delivered on the contract provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass group A inspection.

4.7 Reliability Burn-In/Debugging (See 3.5). Each unit on order shall be subjected to a 48 hour (on time) burn-in period. The power shall be turned on during the period and the console shall be in full operating mode. At the end of the 48 hour period, the operating tests specified in para 3.6 shall be performed. The last 24 hours of burn-in/debugging shall be failure free. The burn-in shall be conducted under the standard conditions of para 4.8. All failures shall be as defined in para 6.6. The contractor may at his option subject the equipment to an additional burn-in/debugging.

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4.7.1 Reports of Failures. Failures during burn-in shall be reported as specified in the contract.

4.8 Standard Conditions. Unless otherwise specified, all inspections shall be performed under the following standard conditions:

a. Power supply voltage: 27.5 ± 0.5Vdc.

b. Warm-up time: Not more than 2 minutes.

c. Temperature: Room ambient (25°C ± 5°C).

d. Humidity: Room ambient temperature up to 90% relative humidity.

e. Altitude: Normal ground level.

f. Vibration: None.

4.9 Electrical Test. (See 3.6).

4.9.1 Operational Data. The following operational data shall be supplied by the contractor to the procuring activity prior to the start of tests:

a. Six copies of test cabling diagram.

b. Electrical Test Plan detaining purpose of test, test limits, and test equipment used.

4.9.2 Electrical Performance. The AN/ASC-15( )(V)1,2 shall meet the electrical specifications and test requirements outlined in para 3.6 of this specification and Section III of TM 11-5821-285-35. Section III, change 1, dated 9 Nov 73, of TM 11-5821-285-35 also applied to this paragraph.

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4.9.3 The AN/ARC-164 installation shall meet the same requirements as those listed in para 5-12, change 1, 9 Nov 73, in TM 11-5821-285-35 except that in para 5-12b(f) the minimum forward power shall be 9 watts instead of 15 watts. See Fig 2 for suggested test setup.

4.9.4 Retransmission of the AN/ASC-15( )(V)1,2 shall be tested as shown in Fig 1.

4.9.4.1 Test Data Collection. Six RF retransmission test frequencies representing the entire 30 MHz to 75.95 MHz band shall be selected as follows:

f_1(MHz)		f_2(MHz)
30	⟷	41
38.5	⟷	50.5
46.5	⟷	58.25
53.75	⟷	66
58.75	⟷	70
63	⟷	74.5

The (dB) attenuation setting necessary to obtain a 10 dB SINAD shall be recorded for each selected RF transmission shown above. A comparison shall then be made between clear text retransmission and secure text retransmission. Clear text attenuation settings for f_1 and f_2 shall not differ by more than 4 dB those of secure test mode retransmission.

4.10 TEMPEST. (See 3.10)

4.10.1 TEMPEST Tests. The communications console shall be tested in accordance with the procedures of Appendix C and the contractor prepared, Government approved, TEMPEST test plan.

4.10.2 Frequency Sampling. TEMPEST tests shall be performed on one first article unit and one production unit selected midway through the production lot.

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4.11 Electromagnetic Interference Test. The console with all equipments installed shall be tested for compliance with the requirements of 3.9 using the test methods of MIL-STD-462, Notice 3, as implemented by a contractor prepared, government approved EMI test plan. Prior to performance of EMI tests, compliance with the bonding requirement of 3.4.2.7.2 shall be ascertained. The data shall be recorded and included in the EMI test report (see 3.9).

4.12 Environmental Tests. Unless otherwise specified, the environmental tests shall be conducted in accordance with the following indicated methods of MIL-STD-810C. When the two documents conflict, this specification shall govern.

4.12.1 Temperature/Altitude Test (See 3.7.1/3.7.2). The console shall be subjected to the test of Method 504.1, Procedure I, Table 504.1-I and 504.1-II, Category 4 of MIL-STD-810C. In addition the test item shall be checked for satisfactory operation as specified in para 3.6 after the end of step 2. At the end of the test, performance testing as specified in para 3.6 shall be made.

4.12.2 Humidity and Rain.

4.12.2.1 Humidity Test (See 3.7.3). The console shall be subjected to test of Method 507.1, Procedure I of MIL-STD-810C. At the end of the tenth cycle, performance testing as specified in para 3.6 shall be made.

4.12.2.2 Rain Test (See 3.7.4). The console shall be subjected to test of Method 506.1, Procedure II of MIL-STD-810C, except step 2, the rate should be 1 $\pm$ 0.5 inches per hour and step 4, the total time should be at least 30 minutes or longer. At the end of the test, the console shall function as specified in para 3.6 of this specification.

4.12.3 Fungus Test (See 3.7.5). The console model in the assembled and as ready for delivery condition, shall be subjected to 28 day test of Method 508.1, MIL-STD-810C. The console test model shall not be specially cleaned for the fungus test, except for the cleaning it receives during or after production. If it is necessary for the manufacturer after assembly to remove accumulated production and handling contaminants by cleaning prior to packaging and delivery, the equipment shall not be tested for three days after cleaning to allow for complete evaporation of the cleaning compound. All enclosed or gasketed assemblies shall be opened and the

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interior sprayed with the specified mixed spore suspension. After 14 and 28 days of test, the control items per paragraphs 3.1.3 and 3.1.4 of Method 508.1, MIL-STD-810C, shall show profuse growth over at least 50% of the area of the control items. After 28 days, based upon visual examination, the equipment assembly shall pass if it shows no more than only sparse microbial growth with restricted tuberal growth development in an area of 1 to 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. No operational test is required after the fungus test to pass the test. The equipment shall fail if it shows more than the growth specified above. The test shall be repeated if the control items fail to show profuse growth after 14 and 28 days of test. The fungus test shall be done before the salt test.

4.12.4 Salt Atmosphere Test (See 3.7.7). The console shall be subjected to the test of Method 509.1, Procedure I, of MIL-STD-810C. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface. (Note: Equipments having moving mechanical parts with exposed bearing surfaces, should have an additional requirement mechanical operation following salt fog testing to insure that no clogging or binding has resulted.)

4.12.5 Explosive Atmosphere Test (See 3.7.8). The console shall be subjected to the test of Method 511.1, Procedure I of MIL-STD-810C. The test altitude shall be 20,000 feet above sea level. Failure criteria shall be in accordance with para 3.3.3a, b, c, and d of the above test procedure. All console mounted equipment shall be installed and operated as specified in para 3.6 and as specified in the above test.

4.12.6 Dust Test (See 3.7.6). The console shall be subjected to the test of Method 510.1, Procedure I, of MIL-STD-810C, with the exception that the air velocity in Steps 1, 2 and 3 shall be 200 + 100 fpm. Step 3 may proceed immediately after stabilization in Step 2 if desired. In step 5, operate the test item as specified in para 3.6 of this specification.

4.12.7 Shock Tests (See 3.7.10). The console shall be subjected to the tests of MIL-STD-810C, Method 516.2, Figure 516.2-2.

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4.12.7.1 Procedure I (Basic Design Test). The console shall be mounted by its normal mounting method in the test stand and shall be operating during the shock test and shall meet the performance requirements specified in para 3.6 after completion of the shock test.

4.12.7.2 Procedure III (Crash Safety Test). The console shall be subjected to the crash safety test while mounted by its normal mounting method. Dummy loads may be used in lieu of installed radios (see 3.7.10.2).

4.12.7.3 Procedure V (Bench Handling Test). The console shall be placed in its normal poistion for servicing on a horizontal solid wooden floor or bench top at least 2 inches thick. At the conclusion of the test, the test item shall be operated as in para 3.6 of this specification (see 3.7.11)

4.12.8 Vibration Tests (See 3.7.9). The console shall be subjected to the test of Method 514.2, Procedure I, Equipment Category C, Table 514.2-III, and Fig 514.2-3 of MIL-STD-810C.

4.12.8.1 Before the test, operate the test item and record preperformance electrical test measurement as specified in para 3.6. Inspect the test item mechanically.

4.12.8.2 Mounting Techniques. The console shall be attached to the vibrator exciter table by its normal mounting means or by means of a rigid fixture capable of transmitting the vibration conditions as specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test set-up. Whenever possible the test load shall be distributed uniformly on the vibrator exciter table in order to reduce to the smallest amount, effects of unbalanced loads. Vibration amplitudes and frequencies shall be measured by techniques that will not significantly affect test item input control or response. The input control sensing device(s) shall be rigidly attached to the vibration table or to the immediate structure, "if used", at or near as possible to the attachement point(s) of the test item.

4.12.8.3 The test shall consist of two parts, Resonance Dwell Test and a Cycling Test. The equipment shall operate satisfactorily without degradation in specified performance of para 3.6 when subjected to the installed equipment portion of both tests.

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4.12.8.3.1 Resonance Dwell. Vibrate the equipment in three mutually perpendicular directions. The dwell frequencies are selected from NOTE 1 of Table 4 and the time schedules from Table 4. The curve selection criteria are taken from Table 5 and the vibration levels and frequency range are selected from Figure 3.

4.12.8.3.2 Cycling. Vibrate the equipment in three mutually perpendicular directions. The time schedules shall be selected from Table 4. The curve selection criteria are taken from Table 5. The vibration levels and frequency range shall be selected from Figure 1.

4.12.8.4 At the end of test, operate the test item and compare electrical test measurements with preperformance measurements. Inspect the test item mechanically.

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4.13 Examination of Product. The equipment configuration shall be inspected to verify that the materials, construction, dimensions, marking, and workmanship, using whatever tools, gauges, scales and the like that are necessary, comply with the requirements of this specification.

4.14 Visual and Mechanical Inspection. The equipment shall be examined for the applicable defects listed in MIL-STD-252. The equipment shall be examined for the following:

a. Workmanship, assembly and fit, mechanical safety, and marking.

b. Materials, parts and finishes.

4.15 Inspection for Interchangeability (See 3.15). The console shall be gaged or measured to determine conformance to the physical interchangeability requirement of para 3.15 when a listed dimension is not within specified or design limits, it shall be considered a major defect.

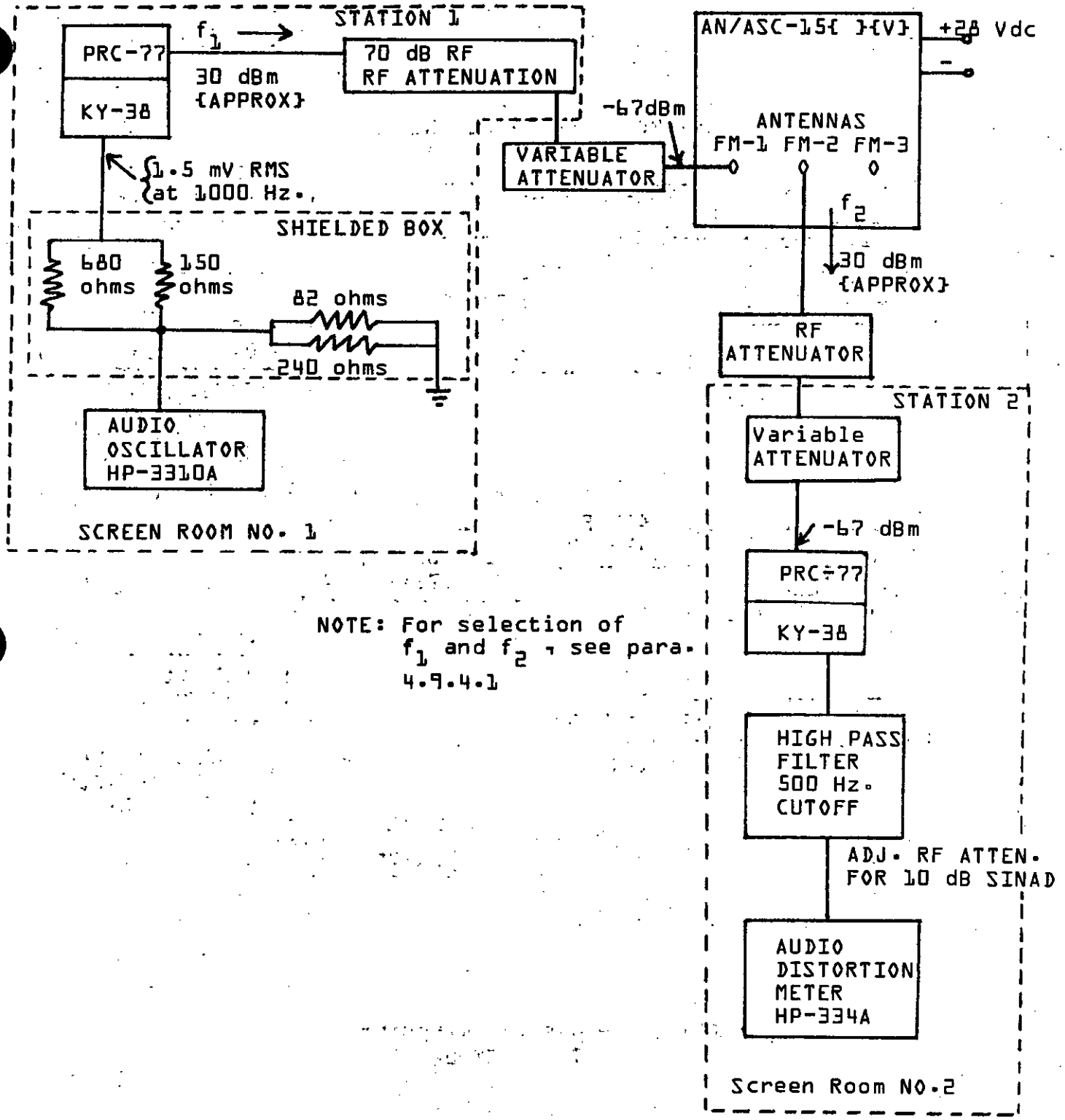
a. External and internal dimensions of console, insertable assemblies, when such dimensions affect rating of parts.

b. Dimensions of cavities, when such dimension affect insertion of item.

c. Location of connectors, locking-pin fasteners, slides and mounting which receive mating parts of plug-in assemblies and major unit, and location of the mating parts on the plug-in assembly or major unit.

4.16 Package Design Validation Test. The contractor shall subject the production unit pack to the first article inspection of MIL-P-116.

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NOTE: For selection of f_1 and f_2, see para. 4.9.4.1

FIGURE 1

SUGGESTED TEST SET UP FOR f_1-f_2 SECURE RETRANSMISSION

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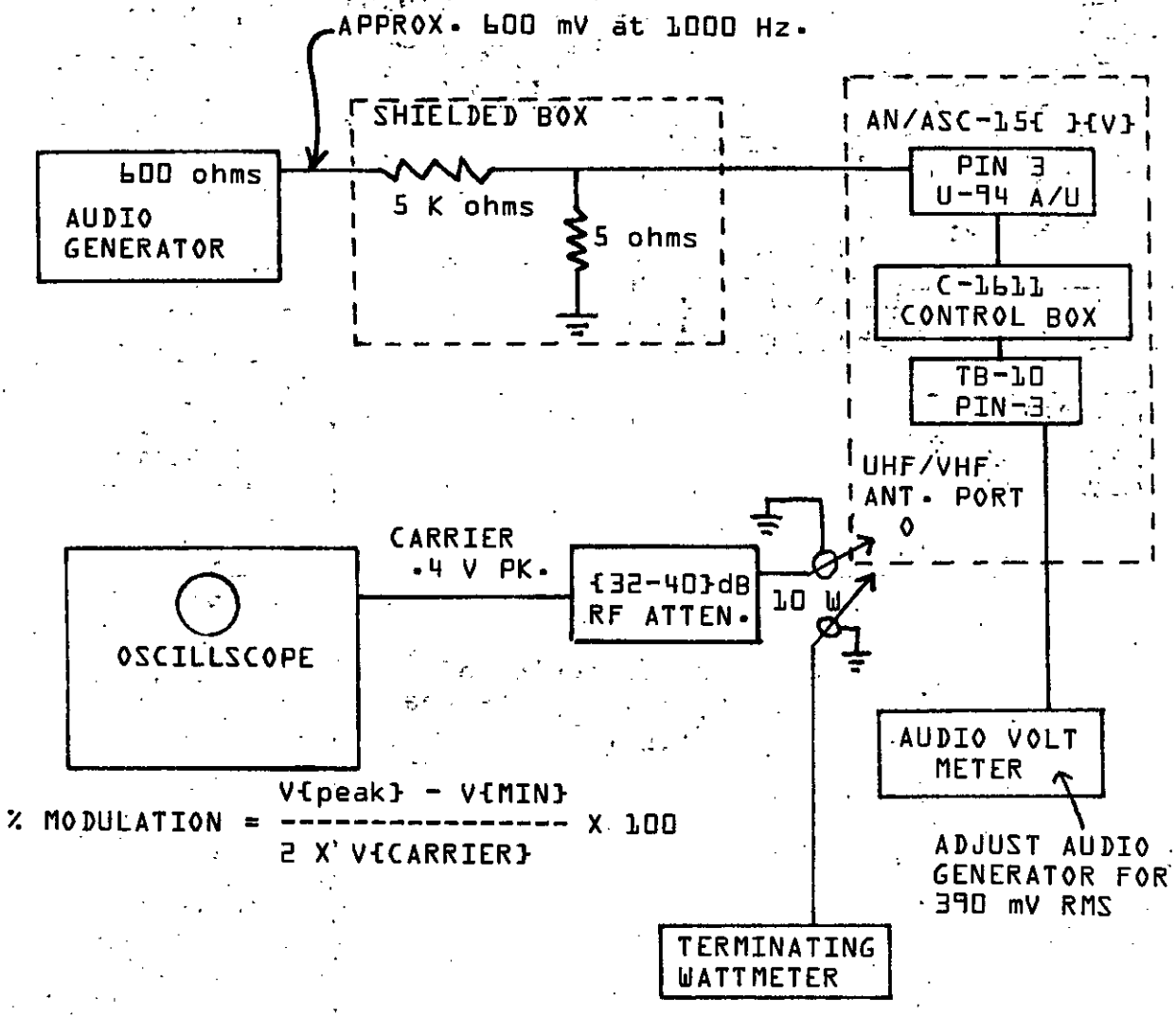


FIGURE 2 - UHF-AM TRANSMITTER
TEST SET-UP

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5. PACKAGING

5.1 Packaging Requirements. The requirements for packaging shall be in accordance with SPI 1G00051.

6. NOTES

6.1 Intended Use. The rack equipment console, when configured with the equipments identified in the appendices, comprises either of two variations of Communications Central AN/ASC-15( )(V) commonly referred to as "console". This console with its associated aircraft installation kit is intended for use in designated UH-1( ) type helicopters as an airborne command post. Two modes of operation exist with each variation, utilizing six operator stations with three receiver-transmitters. One mode provides command tactical communications while the other mode provides secure/non-secure automatic retransmission. Also, in the retransmission mode, the function of command monitoring of an encrypted retransmission message is available.

6.1.1 Communications Central AN/ASC-15( )(V)1. This variation consists of three VHF-FM tactical receiver-transmitters, all of which can be used for envrypted messages with six intercom stations. A receiver-transmitter RT-1167/ARC-164(V) may be installed in lieu of the #3 tactical radio thereby providing a UHF communication capability.

6.1.2 Communications Central AN/ASC-15( )(V)2. This variation consists of two VHF-FM tactical receiver-transmitters, which may be used for encrypted messages, six intercom stations and one Radio Set AN/ARC-51BX. The receiver-transmitter RT-1167/ARC-164(V) may be installed in lieu of the AN/ARC-51BX.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or level B preservation and packing (see Section 5).

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c. Technical literature required.

d. When rough handling test is required (see Section 4).

e. When first article packaging inspection is required.

6.3 Data Items.

a. Test Plan.

b. Packaging.

c. Test Report.

d. Literature and drawing updates.

e. EMI/TEMPEST Test Plan.

f. Preproduction inspection.

g. Place of inspection.

h. Submission of Engineering Change Proposals (ECPs).

6.4 First Article and Group C Inspection.

6.4.1 First Article. The AN/ASC-15( )(V)1,2 shall be tested and approved under the appropriate provision of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a first production item and should consist of a minimum of 4 units. The contractor should follow the specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

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6.4.2 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.6.2.1.6).

6.4.2.1 Initial Shipments. Initial shipment of end items will be contingent only upon successful completion of all Group C tests (see 4.6.2.1).

6.5 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

6.6 Definitions. Reliability failure definitions are as follows:

a. Failure. A failure in the inability of the equipment to perform within specified limits under specified operating conditions.

b. Relevant Failure. All failures shall be considered relevant until proven to the satisfaction of the procuring agency to fulfill the criteria defining a non-relevant failure. Only relevant failures will be used in MTBF calculations. Except as defined in paragraph (b) or when clearly attributed to a condition external to the equipment under test which is not a test requirement and non encountered in service, all failures shall be considered relevant failures.

c. Non-Relevant Failures. The following failures shall be classified as non-relevant (non counted in MTBF calculations), but shall be

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subjected to the same analysis, repair, and corrective action (if appropriate) requirements as relevant failures.

d. Test Equipment Failures. All failures of test equipment and associated fixtures required to conduct testing shall be considered non-relevant.

e. Mechanical Failures. All failures of control knobs, bolts, screws, paint peeling and other mechanical or cosmetic natured items of equipment which do not affect the operation of the equipment to specification limits will be non-relevant.

6.7 Packaging.

6.7.1 Preproduction Pack. When specified in the contract or order one acceptable model of Communications Central AN/ASC-15( )(V)1 and AN/ASC-15( )(V)2 shall be packed in a manner that will pass the first article inspection prescribed in MIL-P-116. Illustrations of the packing procedure, together with Bills of Material, will be prepared and furnished in accordance with the provisions of Government Drawing SC-A-362100.

6.7.2 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in the specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.9 Since this equipment contains communication security equipment, the contractor shall not make a public release, publish photographs or description of this equipment without prior approval of the contracting officer.

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6.10 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example, AN/ASC-15AV1. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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APPENDIX A

Major Component List Comprising Communications Central AN/ASC-15( )(V)1

10. SCOPE

10.1 Scope. This appendix lists the major components of the AN/ASC-15( )(V)1.

20. Components.

a. Three RT-823/ARC-131 Receiver-Transmitters.

b. Six C-1611/AIC Control, Intercommunication Sets.

* c. Three C-7088/ARC-131 Radio Set Controls.

* d. Three C-8157/ARC Control-Indicators (KY-28).

e. Three TSEC/KY-28 Communications Security Encoders.

f. One HYL-3 Regenerative Repeater.

g. One 8-Day Manual Wind Clock, Aircraft Type A-13A-1.

h. Three MT-3664/ARC-131 Mountings.

i. One HYL-3 Shockmount.

j. Three Hardmounts for the TSEC/KY-28.

k. Three Sprague Power Line Filters (28Vdc, 15 amps #JN17-3346A).

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* 1. One RT-1167/ARC-164 Radio Set.

m. Four Headset/Microphone H-157/AIC.

* When the AN/ARC-164 is used, only two C-7088/ARC-131 radio set controls and two C-8157/ARC secure voice control indicators are required.

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APPENDIX B

Major Component List Comprising Communications Central AN/ASC-15( )(V)2

10. SCOPE

10.1 Scope. This appendix lists the major components of the AN/ASC-15( )(V)2.

20. Components.

a. Two RT-823/ARC-131 Receiver-Transmitter Radios.

b. Six C-1611/AIC Control, Intercommunication Sets.

c. Two C-7088/ARC-131 Radio Set Controls.

d. Two C-8157/ARC Control Indicators (KY-28).

e. Two TSEC/KY-28 Communications Security Encoders.

f. One AN/ARC-51BX Receiver-Transmitter Radio.

* g. One C-6287/ARC-51BX Control, Radio Set.

h. One HYL-3 Regenerative Repeater.

i. One 8-Day Manual Wind Clock, Aircraft Type A-13A-1.

j. Two MT-3664/ARC-131 Mountings.

k. One HYL-3 Shockmount.

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l. Two Hardmounts for the TSEC/KY-28.

m. Three Sprague Power Line Filters (28Vdc, 15 amps #JN17-3346A).

* n. One RT-1167/ARC-164 Radio Set.

o. Four Headset/Microphone H-157/AIC.

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* When the AN/ARC-164 is used, no C-6287/ARC-51BX Control, Radio Set is required.

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